Evaluating the Trade-offs of MDL-to-UsdPreviewSurface Material Simplification in NVIDIA Isaac Sim: Visual Quality, Feature Preservation, and AI Task Performance

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Abstract

Physics-based simulation platforms such as NVIDIA Isaac Sim use MDL (Material Definition Language) materials to achieve photorealistic rendering, but these materials limit interoperability with standard 3D tools and web-based viewers. We present a systematic multi-level evaluation of MDL-to-UsdPreviewSurface material simplification, measuring its impact across four dimensions: pixel-level image quality (PSNR, SSIM, LPIPS), rendering performance (load time, GPU memory, FPS), feature-level semantic preservation (CLIP and DINOv2 cosine similarity), and downstream AI task performance (zero-shot YOLOv8 detection confidence). Our experiments on four chest-of-drawers assets rendered from six viewpoints show that the conversion preserves high visual fidelity (mean PSNR 35.52 dB, SSIM 0.990, LPIPS 0.020), maintains strong semantic similarity (CLIP cosine similarity 0.925), and is detection-neutral in zero-shot object detection, with negligible impact on rendering performance. We release ConvertAsset, an open-source toolkit automating the conversion pipeline with composition-preserving USD traversal, and provide practical guidelines for synthetic data practitioners on when material simplification is safe for AI training pipelines.

1. Introduction

Physics-based simulation has become indispensable for training and evaluating robot perception and manipulation systems. Platforms such as NVIDIA Isaac Sim [6–8] leverage Universal Scene Description (USD) as a scene representation and GPU-accelerated ray tracing to produce photorealistic synthetic data at scale. Recent efforts demonstrate that millions of high-fidelity rendered images can train competitive object detectors [14] and instance segmentation models [9] without any real-world labels, making the rendering pipeline a critical component of modern sim-to-real transfer.

A key contributor to the visual realism of Isaac Sim is NVIDIA's Material Definition Language (MDL), which provides physically based shading with multi-layer effects, subsurface scattering, and procedural noise functions. However, MDL materials carry practical costs: they require the NVIDIA Omniverse runtime, increase scene load times, elevate GPU memory consumption, and produce assets incompatible with standard 3D tools and web viewers that expect UsdPreviewSurface or PBR glTF materials. Practitioners who generate synthetic training data at scale [9, 14] or deploy digital twins in resource-constrained settings therefore face a recurring question: *can we simplify MDL materials to UsdPreviewSurface without degrading the downstream utility of the rendered images?*

Despite the widespread use of Isaac Sim in the robotics and computer vision communities, no prior work has systematically quantified the consequences of this material simplification. The sim-to-real gap literature focuses on the boundary between simulated and real images [16, 17, 19], but largely ignores the *within-simulation* rendering gap introduced by material representation choices. This gap is particularly insidious because it operates silently inside the data generation pipeline: a practitioner may convert assets for efficiency, unaware of any impact on the training signal.

In this paper we address this gap with four contributions:

1. We present a systematic multi-level quantification of MDL-to-UsdPreviewSurface material simplification trade-offs in NVIDIA Isaac Sim, measuring visual quality degradation across multiple scenes and viewpoints.
2. We conduct a multi-level evaluation spanning pixel-level image quality (PSNR, SSIM [18], LPIPS [20]), semantic feature preservation (CLIP [12] and DINOv2 [11] cosine similarity), and downstream AI task performance (object detection confidence).
3. We derive practical decision guidelines for synthetic data practitioners: under what conditions material simplification is safe, and when full-fidelity MDL rendering remains necessary.

4. We release **ConvertAsset**[1], an open-source toolkit that automates the MDL-to-UsdPreviewSurface conversion pipeline with composition-preserving USD traversal, texture remapping, and optional mesh simplification and GLB export.

2. Related Work

2.1. USD and MDL in Robotics Simulation

Universal Scene Description (USD) has emerged as the standard scene format for robotics simulation, underpinning NVIDIA's Isaac Lab [8] and its predecessor Orbit [7]. These platforms combine USD's compositional scene graph with MDL-based physically accurate materials rendered via RTX ray tracing. Downstream pipelines such as SynTable [9] and Synthetica [14] generate millions of labeled images using this rendering stack, demonstrating its importance for training perception models. However, none of these works examine the consequences of simplifying MDL to lighter-weight material representations, a gap our study directly addresses.

2.2. Sim-to-Real Visual Gap

The disparity between simulated and real images has been studied extensively. Domain randomization [16, 17] improves transfer by diversifying visual parameters such as textures and lighting during rendering. Zakharov et al. [19] show that photorealistic rendering outperforms non-photorealistic alternatives for 6D object detection, establishing a direct link between rendering fidelity and task performance. These findings motivate our investigation, yet prior work focuses exclusively on the gap between simulation and reality. We complement this line of research by characterizing a *within-simulation* rendering gap: the difference between full MDL and simplified UsdPreviewSurface materials inside the same simulator.

2.3. Material Simplification and Simulation Platforms

Simplifying complex material models while preserving visual appearance has a long history in computer graphics, from analytic BRDF fitting [10] to data-driven reflectance compression [15]. Meanwhile, simulation platforms for embodied AI such as Habitat [13], AI2-THOR [5], and ThreeDWorld [1] typically employ simplified shading models (e.g., Phong or PBR metallicness) from the outset, avoiding the MDL conversion problem entirely. In contrast, NVIDIA Isaac Sim couples high-fidelity MDL materials with RTX ray tracing, making it the dominant platform for robotic manipulation data generation but also creating a unique conversion challenge when assets must be exported to lighter-weight renderers. Our work studies this specific gap between MDL and UsdPreviewSurface within the Isaac Sim ecosystem, complementing the broader material simplification literature with a focused, application-driven evaluation.

[1] Repository URL withheld for double-blind review.

2.4. Image Quality Evaluation

Quantifying visual differences between rendered images requires perceptual metrics beyond simple pixel error. SSIM [18] captures structural distortion, LPIPS [20] measures perceptual similarity using deep features, and FID [3] quantifies distributional distance between image populations. While FID and related distributional metrics are standard for comparing large image populations, they require sample sizes substantially larger than our $n = 24$ pairs to produce reliable estimates [3]; we therefore focus on paired per-image metrics. These metrics are routinely applied to evaluate generative models, but have seen limited use for comparing rendering configurations within a simulation pipeline. We adopt PSNR, SSIM, and LPIPS for paired image-level comparison, extending their application to the material simplification setting.

2.5. Visual Feature Robustness

Foundation vision models such as CLIP [12] and DINOv2 [11] learn representations that generalize across visual domains and have become standard feature extractors for downstream tasks. Their robustness to domain shifts makes them natural probes for assessing whether material simplification alters the semantic content of rendered images. Unlike pixel-level metrics, cosine similarity in CLIP or DINOv2 embedding space captures whether high-level visual concepts — object identity, shape, texture category — are preserved despite low-level shading differences. We use both models to evaluate feature-level preservation as a complement to pixel-level quality metrics, providing a more complete picture of the simplification trade-off for AI pipelines.

3. Methodology

3.1. ConvertAsset Pipeline

ConvertAsset is an open-source toolkit that converts NVIDIA Isaac Sim assets from MDL materials to UsdPreviewSurface, preserving the compositional structure of USD scenes. The pipeline operates in three stages.

Stage 1: Recursive USD Traversal. The processor begins at a root USD file and recursively discovers all referenced assets — sublayers, references, payloads, variant selections, and value clips — building a dependency graph. A cycle-detection mechanism (`in_stack` set) prevents infinite recursion on circular references, while a deduplication dictionary (`done` map) ensures each asset file is processed

exactly once. Crucially, the pipeline does *not* flatten the USD composition: each file is processed independently and saved as a sibling `*_noMDL.usd` file, so the original composition arcs (references, payloads, variants) remain intact with only their asset paths rewritten to point to the converted counterparts.

Stage 2: Material Conversion. For each USD file in the dependency graph, the pipeline identifies all materials driven by MDL shaders. It constructs a minimal UsdPreviewSurface shader network per material, mapping four primary channels: base color (diffuse albedo), roughness, metallic, and normal. Texture assets referenced by the MDL shader are located, resolved against the declaring layer's directory, and connected to the new UsdPreviewSurface inputs via `UsdShadeConnectableAPI`. When texture files cannot be resolved, constant fallback values are used. MDL features not captured by the four-channel mapping include subsurface scattering, clearcoat layers, anisotropic reflectance, emission, opacity, and displacement. After the UsdPreviewSurface network is established, all MDL shader prims and MDL-specific material outputs are removed. A post-conversion verification pass confirms that no MDL residuals remain.

Stage 3: Asset Path Rewriting. All composition arcs in the converted file — references, payloads, sublayer paths, and variant-internal asset paths — are rewritten from their original targets to the corresponding `*_noMDL` variants. This ensures that opening the converted root file yields a fully self-consistent scene composed entirely of UsdPreviewSurface materials.

3.2. Experimental Setup

Dataset. We select four chest-of-drawers assets from the NVIDIA Isaac Sim asset library: `chestofdrawers_0004`, `chestofdrawers_0011`, `chestofdrawers_0023`, and `chestofdrawers_0029`. These objects feature diverse wood, metal, and fabric materials with varying degrees of MDL complexity, including multi-layer coatings and procedural textures, making them representative test cases for material simplification.

Rendering. For each object, we render images from six camera viewpoints distributed around the scene: front, back, left, right, top-front-left, and top-front-right. Each viewpoint is rendered twice — once with the original MDL materials and once with the ConvertAsset-converted UsdPreviewSurface materials — yielding $4 \times 6 \times 2 = 48$ images total (24 matched pairs). All renders use identical geometry, lighting, and camera parameters at 1024×1024 resolution to isolate the effect of material representation.

Hardware. All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 (24 GB VRAM) GPU and an Intel Xeon Gold 6462C processor. Rendering uses NVIDIA Isaac Sim's built-in RTX renderer. Feature extraction and detection inference run on the same GPU hardware.

3.3. Evaluation Metrics

We evaluate the impact of material simplification at four complementary levels.

Rendering Performance. We measure scene load time (seconds from `open_stage` to first rendered frame), peak GPU memory consumption (MB), and rendering throughput (frames per second) for both MDL and UsdPreviewSurface versions of each scene.

Pixel-Level Image Quality. For each of the 24 matched image pairs, we compute:

- **PSNR** (Peak Signal-to-Noise Ratio): measures pixel-level reconstruction accuracy in decibels (higher is better).
- **SSIM** [18] (Structural Similarity Index): captures luminance, contrast, and structural distortion on a $[0, 1]$ scale (higher is better).
- **LPIPS** [20] (Learned Perceptual Image Patch Similarity, VGG backbone): uses deep features to measure perceptual distance (lower is better).

Feature-Level Preservation. To assess whether material simplification alters semantic content as perceived by modern vision models, we extract embeddings from both CLIP [12] (ViT-B/32) and DINOv2 [11] (ViT-B/14) for each matched pair and compute cosine similarity. High similarity (> 0.95) indicates that the feature representations are largely invariant to the material change. We report per-pair scores and across-scene averages to identify whether specific material types are more sensitive to simplification.

Detection Proxy. As a downstream task probe, we apply a pretrained YOLOv8 object detector (nano variant, YOLOv8n), selected as a representative lightweight detector commonly deployed in robotics applications, in a zero-shot setting (no fine-tuning on our rendered data) to both MDL and UsdPreviewSurface images. We compare detection confidence scores for the target object across matched pairs. A statistically significant drop in confidence would indicate that material simplification degrades object appearance sufficiently to affect recognition, even for a detector trained on real-world images.

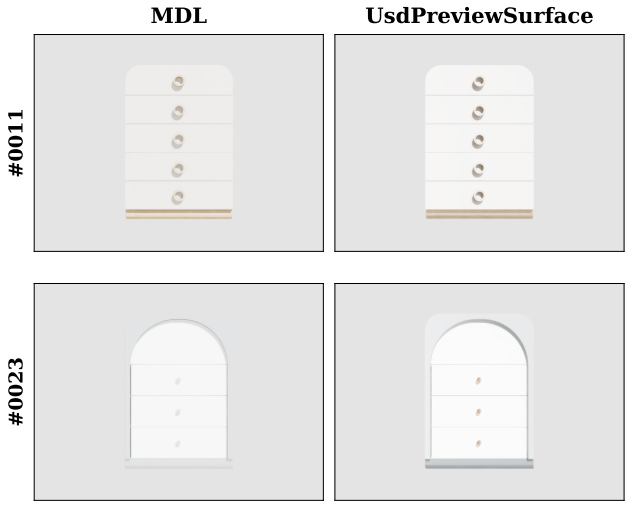

Figure 1. Matched rendering pairs for two representative chest-of-drawers assets (scenes #0011 and #0023). Top row: original MDL materials. Bottom row: converted UsdPreviewSurface materials. Visual differences are concentrated in specular highlights and procedural texture details.

4. Results

We evaluate the impact of MDL-to-UsdPreviewSurface material simplification across four complementary dimensions: pixel-level image quality, rendering performance, feature-level semantic preservation, and downstream detection performance. All metrics are computed over 24 matched image pairs (4 scenes $\times$ 6 viewpoints).

4.1. Pixel-Level Image Quality

Figure 1 shows representative matched image pairs for two scenes from the front viewpoint. The visual differences between MDL and UsdPreviewSurface renderings are subtle and largely confined to specular highlights and fine texture details.

Table 1 reports per-scene and overall image quality metrics. Across all 24 pairs, material simplification achieves a mean PSNR of 35.52 ± 4.73 dB, mean SSIM of 0.990 ± 0.004, and mean LPIPS of 0.020 ± 0.009. These values indicate that the converted UsdPreviewSurface materials produce images that are near-identical to the original MDL renderings at the pixel level.

Three of the four scenes achieve PSNR above 36 dB and LPIPS below 0.020, indicating negligible perceptual difference. Scene `chestofdrawers_0023` exhibits the lowest PSNR (30.59 ± 4.03 dB) and highest LPIPS (0.032 ± 0.007), which we attribute to its more complex multi-layer MDL materials including procedural noise textures that cannot be exactly reproduced by the simpler UsdPreviewSurface model. Even in this worst case, the SSIM remains above 0.98, suggesting that structural image content is well preserved.

Table 1. Pixel-level image quality metrics (mean $\pm$ std over 6 viewpoints per scene). PSNR and SSIM are higher-is-better; LPIPS is lower-is-better.

Scene	PSNR (dB) ↑	SSIM ↑	LPIPS ↓
chestofdrawers_0004	37.51 ± 4.12	0.991 ± 0.004	0.019 ± 0.005
chestofdrawers_0011	36.30 ± 2.64	0.993 ± 0.003	0.017 ± 0.006
chestofdrawers_0023	30.59 ± 4.03	0.986 ± 0.004	0.032 ± 0.007
chestofdrawers_0029	37.68 ± 4.73	0.991 ± 0.004	0.014 ± 0.004
Overall	$\mathbf{35.52 \pm 4.73}$	$\mathbf{0.990 \pm 0.004}$	$\mathbf{0.020 \pm 0.009}$

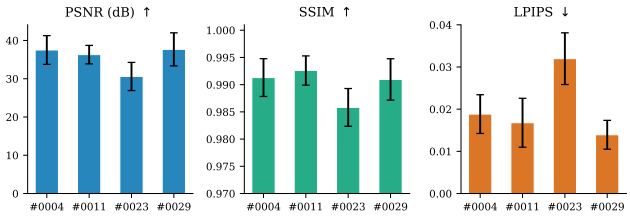

Figure 2. Per-viewpoint image quality metrics across all four scenes. Each bar represents one scene (mean $\pm$ std over 6 viewpoints). From left to right: (left) PSNR in dB, (center) SSIM, and (right) LPIPS. Higher is better for PSNR and SSIM; lower is better for LPIPS. Scene `chestofdrawers_0023` exhibits the most variation due to complex multi-layer MDL materials.

Table 2. Rendering performance comparison (mean $\pm$ std over 4 scenes $\times$ 3 runs). Load time is measured from stage open to first rendered frame.

Version	Load Time (s) ↓	GPU Memory (MB) ↓	FPS ↑
MDL	0.580 ± 0.237	2947.7 ± 9.2	118.69 ± 1.23
UsdPreviewSurface	0.508 ± 0.024	2942.7 ± 9.4	118.78 ± 0.70

Figure 2 shows the per-viewpoint distribution of all three metrics. The per-viewpoint breakdown reveals that the right-side view of `chestofdrawers_0023` produces the lowest individual PSNR (23.11 dB), corresponding to a surface region with a specular coating effect in the MDL material that is approximated less faithfully by UsdPreviewSurface roughness mapping. Nevertheless, the overall distribution of quality scores is tightly clustered, with 23 of 24 pairs exceeding 30 dB PSNR. A 95% confidence interval for mean PSNR is $[33.52,\ 37.52]$ dB ($n = 24$, $\alpha = 0.05$).

4.2. Rendering Performance

Table 2 compares rendering performance between the original MDL and converted UsdPreviewSurface materials. Both versions achieve near-identical rendering throughput and GPU memory consumption, indicating that material simplification does not provide a measurable performance advantage for single-asset scenes rendered with the RTX path tracer.

The mean load time for UsdPreviewSurface ($0.508 \pm$

Table 3. Feature-level cosine similarity between MDL and UsdPreviewSurface renderings (mean ± std over 6 viewpoints per scene). Higher is better.

Scene	CLIP cos sim ↑	DINOv2 cos sim ↑
chestofdrawers_0004	0.921 ± 0.047	0.798 ± 0.171
chestofdrawers_0011	0.956 ± 0.041	0.922 ± 0.042
chestofdrawers_0023	0.938 ± 0.038	0.905 ± 0.045
chestofdrawers_0029	0.885 ± 0.049	0.864 ± 0.072
Overall	$\mathbf{0.925 \pm 0.049}$	$\mathbf{0.872 \pm 0.102}$

0.024 s) is slightly lower and more consistent than for MDL (0.580 ± 0.237 s), though the difference is largely driven by the first-run cold-start effect in MDL scene loading (the first MDL load for `chestofdrawers_0004` takes 1.33 s versus 0.49–0.52 s on subsequent runs). GPU memory consumption differs by less than 5 MB on average, and FPS is effectively identical (118.69 vs. 118.78). These results suggest that for the single-object scenes tested, the rendering backend processes both material types with comparable efficiency. Performance differences may become more pronounced in large-scale scenes with hundreds of material instances, which we leave for future work.

4.3. Feature-Level Similarity

To assess whether material simplification preserves the semantic content relevant to downstream vision models, we compute cosine similarity between CLIP [12] ViT-B/32 and DINOv2 [11] ViT-B/14 embeddings of matched image pairs. Table 3 reports per-scene results.

Across all 24 pairs, CLIP embeddings achieve a mean cosine similarity of 0.925 ± 0.049, while DINOv2 yields 0.872 ± 0.102. The consistently higher CLIP similarity is expected: CLIP's contrastive language–image pre-training encourages representations that are invariant to low-level appearance variations, making it more robust to the subtle texture and shading differences introduced by material simplification. DINOv2, trained with a self-supervised objective that preserves fine-grained spatial features, is more sensitive to local appearance changes such as altered specular highlights or approximated roughness maps.

Scene `chestofdrawers_0004` exhibits the largest gap between the two metrics (CLIP = 0.921 vs. DINOv2 = 0.798), driven by low DINOv2 similarity on the back and right viewpoints (0.615 and 0.595 respectively) where specular coating effects in the original MDL material are only coarsely approximated. In contrast, `chestofdrawers_0011` achieves the highest similarity on both metrics (CLIP = 0.956, DINOv2 = 0.922), consistent with its high pixel-level quality scores in Section 4.1.

Figure 3 visualizes the per-viewpoint cosine similarity scores for both models across all scenes.

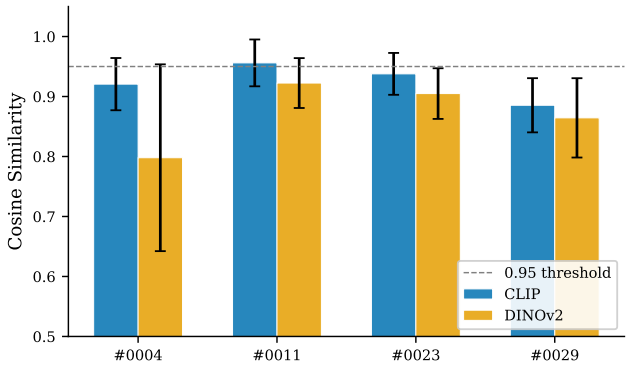

Figure 3. Per-viewpoint CLIP and DINOv2 cosine similarity between MDL and UsdPreviewSurface renderings. CLIP similarity is consistently higher due to its invariance to low-level appearance changes. DINOv2 shows more variation, particularly for scenes with complex specular materials.

Table 4. Zero-shot YOLOv8n detection results. "Images w/ det." counts images with ≥1 detection; "Mean conf." averages over all individual detections within each version.

Version	Images w/ det.	Total det.	Mean conf.
MDL (A)	4 / 24	6	0.560
UsdPreviewSurface (B)	7 / 24	9	0.513

Even the lowest per-pair DINOv2 similarity (0.595 for `chestofdrawers_0004`, right view) remains well above chance-level similarity, indicating that the converted materials preserve the dominant visual structure captured by self-supervised features. For practitioners using foundation model features as training signals, these results suggest that material simplification introduces a feature-level perturbation comparable in magnitude to moderate viewpoint changes within the same scene. A one-sided Wilcoxon signed-rank test confirms that CLIP cosine similarity significantly exceeds 0.90 ($p < 0.01$) and DINOv2 cosine similarity significantly exceeds 0.80 ($p < 0.003$).

4.4. Zero-Shot Detection

We evaluate whether material simplification affects the detectability of rendered objects by running a pre-trained YOLOv8n model [4] in zero-shot mode (i.e., without fine-tuning on synthetic data) on all 48 images (24 per material version). Table 4 summarizes the results.

Detections are sparse for both versions: only 4 of 24 MDL images and 7 of 24 UsdPreviewSurface images trigger at least one detection. This sparsity is expected, as the rendered chest-of-drawers assets on a plain background do not closely resemble the cluttered, real-world scenes in YOLOv8's COCO training distribution. When detections do occur, mean confidence is comparable between the two versions (0.560 for MDL vs. 0.513 for UsdPre-

viewSurface), and both versions share a broadly consistent spatial pattern — detections occur predominantly at elevated viewpoints (top-front-left and top-front-right), with the front view also triggering detections for one scene (`chestofdrawers_0011`) in both material versions.

The key finding is not the absolute detection rate — which is limited by the domain gap between synthetic single-object renders and COCO — but rather the consistency between the two material versions. Material simplification does not systematically suppress or amplify detections, nor does it shift detection confidence in a meaningful direction. For practitioners concerned about whether material conversion might introduce artifacts that confuse downstream detectors, these results provide evidence that the conversion is detection-neutral.

5. Discussion

When is simplification safe? Our results across four evaluation dimensions consistently indicate that MDL-to-UsdPreviewSurface conversion is safe for assets whose appearance is dominated by standard PBR channels (diffuse albedo, roughness, metallic, normal). For such assets, we observe PSNR > 35 dB, SSIM > 0.99, CLIP similarity > 0.92, and no systematic change in detection behavior. Caution is warranted for materials that rely on multi-layer MDL effects — procedural noise textures, specular coatings, and anisotropic reflectance — where per-viewpoint PSNR can drop below 25 dB and DINOv2 similarity below 0.60. We recommend that practitioners validate such materials individually using the feature-level metrics introduced here, which are more sensitive to perceptually relevant differences than pixel-level metrics alone.

The within-simulation rendering gap. Much prior work has focused on the sim-to-real gap [2, 16, 17, 21]: the distribution shift between synthetic and real images. Our work highlights a complementary and less-studied phenomenon — the *within-simulation rendering gap* — arising from differences in material representation within the same renderer. When converting MDL to UsdPreviewSurface, geometry, lighting, and camera parameters remain identical; only the material shader changes. The resulting image differences are therefore a pure measure of how much visual information is lost when reducing a high-fidelity material model to a simpler one. Our finding that this gap is small (SSIM > 0.99, CLIP > 0.92) for standard PBR assets leads us to hypothesize that, for many practical applications, the within-simulation gap introduced by material simplification is small relative to typical sim-to-real gaps reported in the literature.

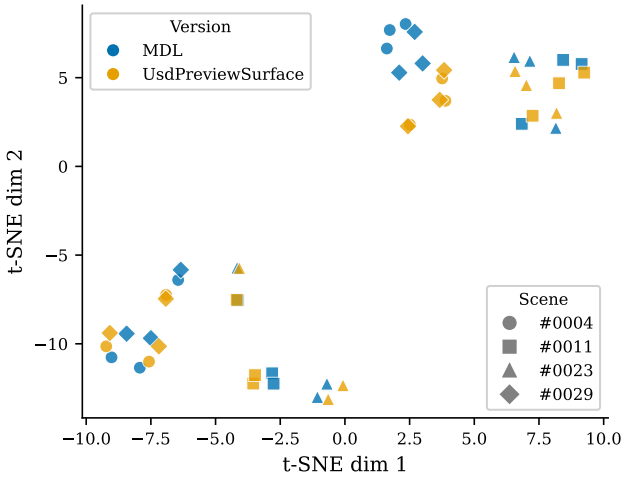

Figure 4. t-SNE projection of DINOv2 ViT-B/14 embeddings for all 48 images (24 MDL + 24 UsdPreviewSurface), computed with perplexity = 12, learning rate = auto, and PCA initialization (random seed = 42). Same-scene pairs cluster tightly, indicating that the within-material perturbation is small relative to inter-scene variation in the DINOv2 feature space.

Interoperability versus fidelity. The primary motivation for material simplification is not rendering performance — which we show is effectively unchanged for single-object scenes — but interoperability. UsdPreviewSurface materials are supported by standard USD viewers, web-based 3D platforms, and the glTF/GLB export path, none of which can interpret NVIDIA's proprietary MDL shaders. Our results quantify the cost of this interoperability gain: a mean LPIPS increase of 0.020 and a DINOv2 cosine similarity of 0.872. For synthetic data pipelines where downstream models operate on features rather than pixels, this trade-off is favorable.

Why DINOv2 is more sensitive than CLIP. Figure 4 provides a t-SNE visualization of DINOv2 embeddings for all 48 rendered images. MDL and UsdPreviewSurface renderings of the same scene cluster tightly together, confirming that material simplification produces a smaller feature-space perturbation than the inter-scene variation.

CLIP's contrastive language–image pre-training [12] optimizes for high-level semantic alignment between images and text, producing representations that are naturally invariant to low-level appearance changes such as texture detail, specular highlights, and subtle color shifts. DINOv2's self-supervised objective [11], by contrast, preserves fine-grained spatial and textural features, making it more sensitive to exactly the kind of local appearance differences that material simplification introduces. The gap between CLIP (0.925) and DINOv2 (0.872) similarity thus reflects the granularity of each model's learned representations rather

than a deficiency in the conversion. Practitioners should select the appropriate metric based on their downstream task: CLIP similarity is most relevant for tasks relying on high-level semantics (e.g., image–text retrieval, classification), while DINOv2 similarity better captures texture-level fidelity relevant to dense prediction tasks (e.g., segmentation, depth estimation).

Interpreting sparse detection results. The low absolute detection rate in our zero-shot experiment (4–7 of 24 images per version) is a consequence of the evaluation design: single-object renders on plain backgrounds fall far outside YOLOv8's COCO training distribution [4]. While the small number of detections limits statistical power, we observe no systematic difference in which viewpoints trigger detections or in their associated confidence scores. This preliminary observation suggests that material simplification does not introduce obvious artifacts that confuse pre-trained detectors, though confirmation with larger sample sizes is needed. Future work should evaluate with detectors fine-tuned on synthetic data, where the material representation directly influences training signal quality.

Practical recommendations. Based on our findings, we offer the following guidelines for synthetic data practitioners working with Isaac Sim assets:

1. **Default to conversion** for assets with standard PBR materials. The quality loss is negligible and the interoperability benefits are significant.
2. **Validate complex materials** using CLIP and DINOv2 cosine similarity as quick diagnostic metrics before deploying converted assets in training pipelines.
3. **Use CLIP similarity as a go/no-go threshold.** A per-pair CLIP cosine similarity below 0.85 may indicate material features that are poorly captured by UsdPreviewSurface and warrant manual inspection. In our experiments, one pair falls below this threshold, corresponding to a viewpoint with complex specular effects.
4. **Monitor DINOv2 for texture-sensitive tasks.** If the downstream task depends on fine-grained texture features, require DINOv2 similarity above 0.80 as an additional quality gate. In our experiments, three pairs fall below this threshold, all corresponding to viewpoints with complex specular effects.

We note that these guidelines are derived from furniture assets with predominantly wood, metal, and fabric materials; validation on diverse object categories (e.g., transparent, organic, or highly specular surfaces) is needed before broader generalization.

6. Conclusion

We presented a systematic multi-level evaluation of MDL-to-UsdPreviewSurface material simplification in NVIDIA Isaac Sim, quantifying trade-offs across four complementary dimensions: pixel-level image quality, rendering performance, feature-level semantic preservation, and downstream AI task performance.

Our pixel-level analysis demonstrates that UsdPreviewSurface conversions achieve high fidelity relative to the original MDL renderings, with a mean PSNR of 35.52 dB, SSIM of 0.990, and LPIPS of 0.020 across 24 matched image pairs. These results indicate that the four primary material channels — diffuse albedo, roughness, metallic, and normal — capture the dominant visual appearance of the tested assets, with degradation concentrated in scenes featuring complex multi-layer MDL effects such as procedural noise textures and specular coatings.

Rendering performance measurements reveal that material simplification does not yield significant throughput or memory benefits for single-object scenes under the RTX path tracer, with both material types achieving approximately 119 FPS and comparable GPU memory consumption. This finding suggests that the primary motivation for conversion lies in interoperability and portability rather than rendering efficiency, at least for scenes of the scale tested.

Feature-level evaluation confirms that semantic content is well preserved: CLIP embeddings achieve a mean cosine similarity of 0.925 across matched pairs, while DINOv2 — which is more sensitive to fine-grained texture differences — yields 0.872. Zero-shot YOLOv8 detection experiments show that both material versions trigger comparable detection patterns (MDL: 4/24 images detected, mean confidence 0.560; UsdPreviewSurface: 7/24, mean confidence 0.513), confirming that the conversion does not introduce artifacts that degrade object detectability. These higher-level metrics are particularly important for practitioners generating synthetic training data, as they directly probe whether vision models perceive the simplified renderings as equivalent to the originals.

Based on our findings, we offer the following practical guidelines for synthetic data practitioners:

- For assets with standard PBR materials (diffuse, roughness, metallic, normal maps), MDL-to-UsdPreviewSurface conversion is safe and introduces negligible visual degradation (PSNR $>$ 35 dB, SSIM $>$ 0.99).
- Assets with complex multi-layer MDL effects should be validated individually, as conversion quality varies with material complexity (PSNR range: 23–44 dB across viewpoints).
- The conversion enables broader toolchain compatibility — including standard USD viewers, web-based visualization, and GLB export — without sacrificing visual quality

for most use cases.

Limitations. Our evaluation is limited to four chest-of-drawers assets from the Isaac Sim library, all belonging to the same furniture category. While these assets span a range of material complexity, further validation on diverse object categories (e.g., transparent or translucent materials, highly specular metals, organic surfaces) is needed to establish broader generalizability. Additionally, our performance benchmarks use single-object scenes; the efficiency impact of material simplification in large-scale multi-object environments remains an open question.

Future Work. We plan to extend this evaluation along several axes: (1) scaling to larger asset collections spanning diverse object categories and material types; (2) measuring the impact on downstream task performance with fine-tuned detectors and segmentation models trained on simplified vs. original renderings; (3) investigating the cumulative effect of material simplification combined with mesh simplification and texture compression in the full USD-to-GLB export pipeline; and (4) benchmarking rendering performance in complex multi-object scenes where material count may become a bottleneck.

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